Cotangent complex

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October 19, 2026

Abstract

The cotangent complex $\mathbb{L}_{B/A}$ is the derived version of the module of Kähler differentials. We first recall why $\Omega_{B/A}$ is the universal linearization of an algebra, then define $\mathbb{L}_{B/A}$ by deriving that construction, and characterize it by the deformation problem it represents. The rest of the note computes: a point in $\mathbb{A}^n$, regular embeddings, the canonical identification $H_1(\mathbb{L}) \simeq I/I^2$ and the gluing rule it implies, what fails without regularity, and a singular point, where $\mathbb{L}$ sees a deformation that Ω^1 misses.

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1 Kähler differentials

Recall that a tangent vector on a scheme X/k is, using a square-zero element ϵ,

$$v \in \mathrm{Hom}_{\mathbf{Sch}/k}(\mathrm{Spec}(k \oplus k\epsilon), X). \tag{1.1}$$

If one wants to fix a base point of the vector v, i.e. $v \in$ "T_pX" where $p \in \mathrm{Hom}_{\mathbf{Sch}/k}(\mathrm{Spec}\, k, X)$, we draw a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}\, k & & \\ \downarrow & \searrow^{p} & \\ \mathrm{Spec}(k \oplus k\epsilon) & \xrightarrow[v]{} & X \end{array} \tag{1.2}$$

Intuitively, v is an infinitesimal deformation of the point p. Following the same philosophy, a "tangent vector field" over a general affine scheme $\operatorname{Spec} A$ is an infinitesimal deformation of the identity:

$$\begin{array}{ccc} \operatorname{Spec} A & & \\ \downarrow & \searrow^{\mathrm{id}} & \\ \operatorname{Spec}(A \oplus M) & \longrightarrow & \operatorname{Spec} A \end{array} \qquad 0 \to M \to A \oplus M \to A \to 0 \tag{1.3}$$

where $A \oplus M$ is a square-zero extension of A. If M has more than one generator, such a diagram specifies several vector fields on $\operatorname{Spec} A$. Now, since tangent vectors are infinitesimal objects, we should expect the existence of a universal augmentation L_A over $\operatorname{Spec} A$ which contains linearized (infinitesimal) information of all possible deformations of $\mathrm{id}_{\operatorname{Spec} A}$. Its function ring will be a square-zero extension of A, which we denote as $A \oplus \Omega_A$.

$$\begin{aligned} \operatorname{Hom}_{\operatorname{Spec} A \backslash \mathbf{Sch}}(\operatorname{Spec}(A \oplus M), \operatorname{Spec} A) &= \operatorname{Hom}_{\operatorname{Spec} A \backslash \mathbf{Sch}/\operatorname{Spec} A}(\operatorname{Spec}(A \oplus M), L_A) && \text{(1.4a)} \\ = \operatorname{Hom}_{\mathbf{CRing}/A}(A, A \oplus M) &= \operatorname{Hom}_{A \backslash \mathbf{CRing}/A}(A \oplus \Omega_A, A \oplus M) && \text{(1.4b)} \\ = \operatorname{Der}(A, M) &= \operatorname{Hom}_{A\mathbf{Mod}}(\Omega_A, M) && \text{(1.4c)} \end{aligned}$$

Here $A\backslash\mathbf{CRing}/A$ denotes augmented A-algebras; the square-zero extensions $A \oplus M$ form a full subcategory, equivalent to $\mathbf{Ab}(\mathbf{CRing}/A)$ by (1.7); morphisms must preserve the section $A \to A \oplus N$. In $\mathbf{CRing}/A$ alone one would get $\operatorname{Hom}_{\mathbf{CRing}/A}(A \oplus \Omega_A, A \oplus M) = \operatorname{Der}(A, M) \times \operatorname{Hom}_A(\Omega_A, M)$. Notation: for an object X of a category $\mathcal{C}$, we write $\mathcal{C}/X$ for objects over X and $X\backslash\mathcal{C}$ for objects under X; thus $\operatorname{Spec} A\backslash\mathbf{Sch}$ corresponds to $\mathbf{CRing}/A$ under Spec, and $\operatorname{Spec} A\backslash\mathbf{Sch}/\operatorname{Spec} A$ to $A\backslash\mathbf{CRing}/A$. Also, $L_A = \operatorname{Spec}(A \oplus \Omega_A)$ is the first-order neighborhood of the zero section of the tangent "bundle" $\operatorname{Spec} \operatorname{Sym}_A \Omega_A$; since the sources $\operatorname{Spec}(A \oplus M)$ are square-zero, maps into either give the same set. Comparing the left-hand side of Eq. (1.4b) and the right-hand side of Eq. (1.4c), it is clear that the functor $A \mapsto \Omega_A$ ought to be defined as the left adjoint to the functor

$$\mathcal{F} : A\mathbf{Mod} \longrightarrow \mathbf{CRing}/A, \qquad M \mapsto A \oplus M. \tag{1.5}$$

A standard construction is given by the conormal module of the diagonal, $\Omega_A = I/I^2$,

$$0 \to I/I^2 \to (A \otimes A)/I^2 \xrightarrow{m} A \to 0, \tag{1.6}$$

where $I = \ker(A \otimes A \xrightarrow{m} A)$ is the ideal of the diagonal $\operatorname{Spec} A \to \operatorname{Spec} A \times \operatorname{Spec} A$.

A special feature of the functor $\mathcal{F}$ is that it lands on abelian group objects as an equivalence of categories.

$$\mathcal{F} : A\mathbf{Mod} \xrightarrow{\sim} \mathbf{Ab}(\mathbf{CRing}/A) \hookrightarrow \mathbf{CRing}/A \tag{1.7}$$

Let us briefly digress to this observation. The unit axiom for an abelian group object N in $\mathbf{CRing}/A$ is the commutative diagrams [in $\mathbf{CRing}$]

$$\begin{array}{ccc} A & \xrightarrow{u} & N \\ {\scriptstyle 1_A}\downarrow & \swarrow & \\ A & & \end{array} \qquad \begin{array}{ccccc} & & \overset{\mathrm{id}}{\frown} & & \\ N = A \times_A N & \xrightarrow{u \times \mathrm{id}} & N \times_A N & \xrightarrow{m} & N \\ & \searrow & \downarrow & \swarrow & \\ & & A & & \end{array} \tag{1.8}$$

The first diagram in Eq. (1.8) indicates that $N = A \oplus M$ has a decomposition as an abelian group. Under this decomposition, the triangle reduces to

$$A \xrightarrow{(\mathrm{id},0)} A \oplus M \xrightarrow{\mathrm{pr}_1} A. \tag{1.9}$$

On the other hand, the pullback square defining $N \times_A N$ with $N = A \oplus M$ reveals that

$$N \times_A N = A \oplus M \oplus M \tag{1.10}$$

where $((a,x),(a,y))$ is identified with (a,x,y). Furthermore, $(0,x,0)\cdot(0,0,y) = 0$, $\forall x, y \in M$. Then tracking the image of $(a,x) \in N = A \oplus M$ along the second diagram in Eq. (1.8) yields

$$(a,x) = (a,(a,x)) \xmapsto{u\times\mathrm{id}} ((a,0),(a,x)) = (a,0,x) \xmapsto{m} (a,x). \tag{1.11}$$

Now, the following computation reveals that $N = A \oplus M$ is a square-zero extension of A.

$$m(a,x,y) = m(a,x,0) + m(0,0,y) = (a,x) + (0,y) = (a,x+y), \tag{1.12a}$$
$$0 = m(0,0,0) = m(0,x,0)\cdot m(0,0,y) = (0,x)\cdot(0,y). \tag{1.12b}$$

The upshot is that, identifying $A\mathbf{Mod} \simeq \mathbf{Ab}(\mathbf{CRing}/A)$, the Kähler differential is given by $\Omega_A = \mathrm{Ab}_A(A \xrightarrow{\mathrm{id}} A)$ where Ab_A is left adjoint to the forgetful functor

$$\mathbf{Ab}(\mathbf{CRing}/A) \to \mathbf{CRing}/A. \tag{1.13}$$

Interpretation: Taking the abelian group objects out of a category $\mathcal{C}/A$ [of functions] amounts to linearizing the category $\mathcal{C}^{\mathrm{op}}$ [of spaces], and the Kähler differential Ω_A is the universal linearization of A.

Relative formulation. Relativization of the absolute Kähler differential $\Omega_A = \Omega_{A/\mathbb{Z}} \to \Omega_{A/R}$ is achieved by repeating the discussion in the category $R\mathbf{CAlg}$.

$$\mathrm{Hom}_{R\mathbf{CAlg}/A}(A, A\oplus M) \simeq \mathrm{Der}_R(A,M) \simeq \mathrm{Hom}_{A\mathbf{Mod}}(\Omega_{A/R}, M) \tag{1.14}$$

Given a morphism $A \to B$ in $R\mathbf{CAlg}$ and a B-module M, we see that the abelianization functor [left adjoint to forgetful functor]

$$\mathrm{Ab}^R_B : R\mathbf{CAlg}/B \to \mathbf{Ab}(R\mathbf{CAlg}/B) \simeq B\mathbf{Mod} \tag{1.15}$$

sends $A \to B$ to $B \otimes_A \Omega_{A/R}$.

$$\mathrm{Hom}_{R\mathbf{CAlg}/B}(A, B\oplus M) \simeq \mathrm{Der}_R(A,M) \simeq \mathrm{Hom}_{A\mathbf{Mod}}(\Omega_{A/R}, M) \simeq \mathrm{Hom}_{B\mathbf{Mod}}(B\otimes_A \Omega_{A/R}, M) \tag{1.16}$$

2 The cotangent complex

2.1 André–Quillen cohomology

It had been known before the advent of cotangent complex in its full generality that commutative rings possess a cohomology theory in the following sense. Let $A \to B \to C$ be morphisms of commutative rings and M be a C-module. Then we have an exact sequence

$$\begin{aligned} 0 &\to \mathrm{Der}_B(C,M) \to \mathrm{Der}_A(C,M) \to \mathrm{Der}_A(B,M) \\ &\to \mathrm{Exalcomm}_B(C,M) \to \mathrm{Exalcomm}_A(C,M) \to \mathrm{Exalcomm}_A(B,M) \end{aligned} \tag{2.1}$$

As we have seen in Eq. (1.4c), the group $\mathrm{Der}_A(B,M)$ classifies the A-algebra sections $B \to B\oplus M$ of the trivial (*split*) square-zero extension of B by M. In contrast, the group $\mathrm{Exalcomm}_A(B,M)$ classifies general infinitesimal algebra extensions $0 \to M \to E \to B \to 0$; see e.g. [GW23]. Note that the first row of Eq. (2.1) is obtained by applying $\mathrm{Hom}(-,M)$ to the sequence

$$C \otimes_B \Omega_{B/A} \to \Omega_{C/A} \to \Omega_{C/B} \to 0. \tag{2.2}$$

The lesson is that while $\Omega_{B/A}$ itself fails to contain the full information about deformations, it might be remedied by incorporating higher-degree data analogous to the Ext functor [derived functor of Hom]. The *cotangent complex* realizes exactly this idea.

2.2 Definition

Recall that in ordinary homological algebra, the groups $\mathrm{Ext}^i_A(B,M)$ are obtained by replacing B by its projective resolution $P_\bullet \to B$ in the category $A\mathbf{Mod}$, applying the functor $\mathrm{Hom}_A(-,M)$ degreewise, and taking cohomology. Since we are working with commutative algebras, we do the same task in the category $A\mathbf{CAlg}$, with the functor $\Omega_{\cdot/A}$ in place of $\mathrm{Hom}_A(-,M)$. However, we do not take cohomology hastily, but keep track of all the information.

Definition 2.1. Let $A \to B$ be a map of commutative rings and $P_\bullet \xrightarrow{\sim} B$ a free simplicial resolution in $A\mathbf{CAlg}$. The cotangent complex of B over A is the simplicial B-module

$$\mathbb{L}_{B/A} := B \otimes_{P_\bullet} \Omega_{P_\bullet/A}, \tag{2.3}$$

regarded as a connective chain complex of B-modules via the Dold–Kan correspondence (§2.5).

A rationale for the definition in Eq. (2.3) is that since $\Omega_{B/A}$ is the abelianization of the arrow $B \xrightarrow{\mathrm{id}} B$ in the category $A\mathbf{CAlg}/B$, we take the cofibrant replacement $P_\bullet \to B$ of $B \xrightarrow{\mathrm{id}} B$ and then abelianize, which leads to the left-derived functor $\mathbb{L}_{B/A}$.

Above construction naturally fits into the general philosophy of derived algebraic geometry, where the objects of interest are defined only up to a homotopy type. The information content of the original definition is not lost, since the concrete homotopy data are kept along [exactly how homotopy equivalent objects are equivalent]. This is illustrated by our choice not to take cohomology but keep the chain complex structure in Eq. (2.3).

How and why the notion of homotopy is engaged in defining a geometric object as a chain complex, rather than simply a module, is clarified by the Dold–Kan correspondence. Note that $\mathbb{L}_{B/A}$ is *not* an algebra: it is a simplicial B-module, the abelianization lands in $\mathbf{Ab}(A\mathbf{CAlg}/B) \simeq B\mathbf{Mod}$ by Eq. (1.15). The algebra in the construction is the resolution $P_\bullet$, a simplicial commutative A-algebra. Dold–Kan identifies simplicial B-modules with connective chain complexes of B-modules, and in characteristic 0 simplicial commutative A-algebras may be modeled by connective commutative differential graded A-algebras (CDGAs) [Qui70]; see §2.5.

2.3 Universal property

Deriving the adjunction (1.16) gives the property that characterizes $\mathbb{L}_{B/A}$ without reference to any resolution: for every B-module M and $n \geq 0$,

$$\mathrm{Map}_{\mathrm{s}A\mathbf{CAlg}/B}\left(B,\ B \oplus M[n]\right) \simeq \mathrm{Map}_B\left(\mathbb{L}_{B/A},\ M[n]\right), \tag{2.4}$$

where $B \oplus M[n]$ is the trivial square-zero extension with M placed in homological degree n. Taking π_0 for $n = 0, 1$ recovers the classical groups of Eq. (2.1) [Qui70; Ill71]:

$$\mathrm{Ext}^0_B(\mathbb{L}_{B/A}, M) = \mathrm{Der}_A(B,M), \qquad \mathrm{Ext}^1_B(\mathbb{L}_{B/A}, M) = \mathrm{Exalcomm}_A(B,M). \tag{2.5}$$

This is the precise sense in which $\mathbb{L}_{B/A}$ "contains the full information about deformations": the sequence (2.1) is the beginning of the long exact sequence obtained by applying $\mathrm{Hom}(-,M)$ to the transitivity triangle (2.6) below.

2.4 Basic properties

We collect, without proof, the properties used below [Ill71; Iye07].

1. *Independence of the resolution.* Any two free resolutions of B are related by a map over B which is unique up to homotopy; the induced map on (2.3) is a quasi-isomorphism. Hence $\mathbb{L}_{B/A}$ is well defined up to canonical quasi-isomorphism.

2. *Degree* 0. $\mathbb{L}_{B/A}$ is connective and $H_0(\mathbb{L}_{B/A}) = \Omega_{B/A}$.

3. *Smooth algebras.* If B is smooth over A (e.g. a polynomial ring), then $\mathbb{L}_{B/A} \simeq \Omega_{B/A}$ in degree 0.

4. *Transitivity.* For $A \to B \to C$ there is a distinguished triangle of C-modules

$$C \otimes_B \mathbb{L}_{B/A} \to \mathbb{L}_{C/A} \to \mathbb{L}_{C/B} \xrightarrow{+1}, \tag{2.6}$$

   whose H_0 is the sequence (2.2).

5. *Localization.* For $g \in A$, $\mathbb{L}_{B_g/A_g} \simeq \mathbb{L}_{B/A} \otimes_B B_g$. Consequently, for a morphism of schemes $Z \to X$ the local cotangent complexes form a complex of sheaves $\mathbb{L}_{Z/X}$ on Z, defined without choosing charts.

2.5 Simplicial commutative rings and CDGA models

Having a simplicial commutative ring $\mathsf{A} = (\mathsf{A}_n)_{n\geq 0} \in \mathrm{s}\mathbf{CRing}$, we define the category $\mathrm{s}\mathbf{AMod}$ of simplicial A-modules as the subcategory of $\mathrm{s}\mathbf{Ab}$ containing those simplicial abelian groups $(M_n)_{n\geq 0}$ such that

1. Each M_n is an A_n-module,

2. Every simplicial map $M_n \to M_k$ is an A_n-module homomorphism.

This means that $M_f(am) = \mathsf{A}_f(a)M_f(m)$, $\forall a \in \mathsf{A}_n$, $\forall m \in M_n$, $\forall f \in \mathrm{Hom}_\Delta([k],[n])$. Every simplicial commutative ring A is a simplicial module over itself, as expected.

Dold–Kan. The normalized chain complex functor N gives an equivalence between simplicial abelian groups and connective chain complexes, $(NM)_n = \bigcap_{i<n} \ker d_i$ with differential $(-1)^n d_n$, and $\pi_n(M) = H_n(NM)$. It upgrades to simplicial modules over a ring B; this is how $\mathbb{L}_{B/A}$ is read as a chain complex.

Free resolutions exist but are large. Every A-algebra B has a canonical free simplicial resolution: apply the free-algebra comonad $\mathrm{Sym}_A(\cdot)$ repeatedly ("polynomials on the elements of B, then on the relations among those, ..."). It is functorial in B, which gives the properties in §2.4, but it is far too large to compute with.

CDGA models. In characteristic 0, one may instead use a *semi-free* connective CDGA $K \xrightarrow{\sim} B$: a free graded-commutative A-algebra on generators in degrees ≥ 0, with a differential satisfying the Leibniz rule. Then $\mathbb{L}_{B/A} \simeq B \otimes_K \Omega_{K/A}$, and $\Omega_{K/A}$ is the free K-module on $\{\mathrm{d}x\}$, one $\mathrm{d}x$ for each generator x, in the degree of x. All computations below are of this kind; the results of §4 hold in every characteristic [Ill71].

3 A point in affine space

Let $R = k[x_1, \ldots, x_n]$ and consider the origin $k = R/(x_1, \ldots, x_n)$. The Koszul complex, so far used as a resolution of k by *R-modules*, is in fact a resolution by an *R-algebra*:

$$K = R[\varepsilon_1, \ldots, \varepsilon_n], \qquad |\varepsilon_i| = 1, \qquad \varepsilon_i\varepsilon_j = -\varepsilon_j\varepsilon_i, \qquad d\varepsilon_i = x_i, \tag{3.1}$$

with d extended by the Leibniz rule. As a complex, $K = \bigwedge^\bullet V$ with $V = \bigoplus_i R\varepsilon_i$ and $d = \iota_x$. It is semi-free and $K \xrightarrow{\sim} k$ because $(x_1, \dots, x_n)$ is a regular sequence.

$$K = \left(\begin{array}{ccccccccc} \bigwedge^n V & \xrightarrow{\iota_x} & \cdots & \xrightarrow{\iota_x} & \bigwedge^2 V & \xrightarrow{\iota_x} & V & \xrightarrow{\iota_x} & R \\ & & & & & & \| & & \\ & & & & & & \bigoplus_{i=1}^n R\varepsilon_i & & \end{array} \right) \xrightarrow{\sim} k, \tag{3.2a}$$

$$\rightsquigarrow \quad \Omega^1_{K/R} = \left(\begin{array}{ccccccccc} \bigwedge^n V \otimes W & \xrightarrow{\iota_x} & \cdots & \xrightarrow{\iota_x} & V \otimes W & \xrightarrow{\iota_x} & W & \longrightarrow & 0 \\ & & & & & & \| & & \\ & & & & & & \bigoplus_{i=1}^n R\,\mathrm{d}\varepsilon_i & & \end{array} \right) = \bigoplus_{i=1}^n K\,\mathrm{d}\varepsilon_i \tag{3.2b}$$

$$\overset{-\otimes_K k}{\rightsquigarrow} \quad \mathbb{L}_{k/R} = \bigoplus_{i=1}^n k\,\mathrm{d}\varepsilon_i \ \ (\text{degree } 1) \ = \ k^n[1]. \tag{3.2c}$$

Here the Kähler differential d has degree 0, $|\mathrm{d}x| = |x|$, and the differential of Ω^1_K is determined by $d\,\mathrm{d} = \mathrm{d}\,d$ and the Leibniz rule. (If d is instead given odd degree, as for the de Rham differential on $\Omega^\bullet_K$ or Connes' B, the same condition reads $d\,\mathrm{d} + \mathrm{d}\,d = 0$; the two conventions differ by the sign $(-1)^{\deg}$.) In particular $\mathrm{d}(d\varepsilon_i) = \mathrm{d}x_i = 0$ since $x_i \in R$, so the differential of $\Omega^1_{K/R} = \bigoplus_i K\,\mathrm{d}\varepsilon_i$ is $\iota_x \otimes 1$, that of K on each summand. Since $K \to k$ kills every ε_i, $\bigwedge^p V \otimes_K k = 0$ for $p \geq 1$, and only the degree-1 generators $\mathrm{d}\varepsilon_i$ survive.

Remark 3.1 (relation to the derived intersection)**.** Tensoring the same resolution with k gives $k \otimes^{\mathbb{L}}_R k = K \otimes_R k = k[\varepsilon_1, \dots, \varepsilon_n]$ with zero differential, the free graded-commutative algebra on the degree-1 generators ε_i. Thus

$$k \otimes^{\mathbb{L}}_R k = \bigwedge^\bullet k^n = \mathrm{Sym}_k(\mathbb{L}_{k/R}) : \tag{3.3}$$

the derived self-intersection of the point is freely generated by its cotangent complex.

Remark 3.2 (absolute check)**.** Over k the generators of K are x_i and ε_i, so $\Omega^1_{K/k} = \bigoplus_i K\,\mathrm{d}x_i \oplus \bigoplus_i K\,\mathrm{d}\varepsilon_i$ with $\mathrm{d}\varepsilon_i \mapsto \mathrm{d}x_i$. After $\otimes_K k$ this is the isomorphism $k^n \xrightarrow{\sim} k^n$, hence acyclic: $\mathbb{L}_{k/k} = 0$, as it must be. Equivalently, in the transitivity triangle (2.6) for $k \to R \to k$, the connecting map $\mathbb{L}_{k/R} = k^n[1] \to \mathbb{L}_{R/k} \otimes_R k[1] = \bigoplus k\,\mathrm{d}x_i[1]$ is $\mathrm{d}\varepsilon_i \mapsto \mathrm{d}x_i$, an isomorphism.

4 Regular embeddings

4.1 The local computation

Let A be a commutative ring and $I = (f_1, \dots, f_c)$ an ideal generated by a regular sequence. The Koszul algebra $K = A[\varepsilon_1, \dots, \varepsilon_c]$, $d\varepsilon_i = f_i$, is again semi-free, and $K \xrightarrow{\sim} A/I$ precisely because f is regular, i.e. $H_{\geq 1}(f; A) = 0$. The computation of §3 goes through verbatim. Writing $V = \bigoplus_i A\varepsilon_i$ and $W = \bigoplus_i A\,\mathrm{d}\varepsilon_i$,

$$\Omega^1_{K/A} = \Big(\bigwedge^c V \otimes W \longrightarrow \cdots \longrightarrow V \otimes W \xrightarrow{\iota_f \otimes 1} W \longrightarrow 0 \Big) = \bigoplus_{i=1}^c K\,\mathrm{d}\varepsilon_i, \tag{4.1a}$$

$$\overset{-\otimes_K A/I}{\rightsquigarrow} \quad \mathbb{L}_{(A/I)/A} = \bigoplus_{i=1}^c (A/I)\,\mathrm{d}\varepsilon_i \ \ (\text{degree } 1). \tag{4.1b}$$

Regularity was used exactly once: to know that K may be used to compute $\mathbb{L}$. The result still depends on the chosen generators f; the next subsection removes this dependence.

4.2 $H_1(\mathbb{L}) \simeq I/I^2$ canonically

Proposition 4.1. *Let $A \to B = A/I$ be surjective, with no assumption on I. Then there is an isomorphism of B-modules*

$$H_1(\mathbb{L}_{B/A}) \simeq I/I^2, \tag{4.2}$$

natural in $A \to B$ and compatible with localization.

Proof. By (2.5) it suffices to compute $\mathrm{Der}_A(B, M)$ and $\mathrm{Exalcomm}_A(B, M)$ for a B-module M.

1. $\mathrm{Der}_A(B, M) = 0$, since a derivation vanishes on the image of A and $A \to B$ is surjective.
2. $\mathrm{Exalcomm}_A(B, M) \simeq \mathrm{Hom}_B(I/I^2, M)$. Given an extension $0 \to M \to E \to B \to 0$ of A-algebras, the structure map $A \to E$ sends I into $\ker(E \to B) = M$, and I^2 into $M \cdot M = 0$; this gives a B-linear map $I/I^2 \to M$. Since $E = \mathrm{im}(A) + M$, the extension is recovered as $E = (A \oplus M)/\{(i, -\psi(i)) : i \in I\}$, i.e. as the pushout of $0 \to I/I^2 \to A/I^2 \to A/I \to 0$ along ψ. The two constructions are inverse to each other. No generators of I are chosen.

Since $\mathbb{L}_{B/A}$ is connective with $H_0 = \Omega_{B/A} = 0$, its lowest homology is in degree 1, and $\mathrm{Ext}^1_B(\mathbb{L}_{B/A}, M) = \mathrm{Hom}_B(H_1(\mathbb{L}_{B/A}), M)$. Combining with item 2, $\mathrm{Hom}_B(H_1(\mathbb{L}_{B/A}), M) \simeq \mathrm{Hom}_B(I/I^2, M)$ naturally in M, and the Yoneda lemma gives $H_1(\mathbb{L}_{B/A}) \simeq I/I^2$. Both sides localize, and so does the isomorphism, since it was built from (2.4) without choices. □

Lemma 4.2 (the isomorphism in coordinates)**.** *In the Koszul model (4.1b), the isomorphism of Proposition 4.1 is $\mathrm{d}\varepsilon_i \mapsto \bar{f}_i$.*

Proof. An element of $\mathrm{Hom}_B(\mathbb{L}_{B/A}, M[1])$ is given in the model by values $m_i \in M$ on the generators $\mathrm{d}\varepsilon_i$, i.e. by an A-derivation $D : K \to M$ with $D(\varepsilon_i) = m_i$. The corresponding extension is obtained by deforming the differential of the resolution: on the algebra $K \oplus M$ (M in degree 0, square-zero) put $d_D\varepsilon_i = f_i - m_i$, and set $E = H_0(K \oplus M, d_D)$. The ideal generated by the $f_i - m_i$ consists of the elements $(\sum_i a_i f_i, -\sum_i \bar{a}_i m_i)$, $a_i \in A$, where $\bar{a}_i$ acts on M through B. If $\sum_i a_i f_i = 0$ then (a_i) is a Koszul relation because f is regular, so every $a_i \in I$ and $\sum_i \bar{a}_i m_i = 0$; hence $M \hookrightarrow E$ and $0 \to M \to E \to B \to 0$ is an extension. In E we have $\sum_i a_i f_i = \sum_i \bar{a}_i m_i$, so the map $I/I^2 \to M$ of item 2 is $\bar{f}_i \mapsto m_i$. Thus the functional $\mathrm{d}\varepsilon_i \mapsto m_i$ on $\mathbb{L}$ corresponds to the functional $\bar{f}_i \mapsto m_i$ on I/I^2, for every M, which is the claim. □

4.3 Globalization and the gluing rule

Corollary 4.3. *Let $Z \subset X$ be a regular embedding with ideal sheaf $\mathcal{I}$ and conormal sheaf $N^\vee_{Z/X} = \mathcal{I}/\mathcal{I}^2$. Then*

$$\mathbb{L}_{Z/X} \simeq N^\vee_{Z/X}[1]. \tag{4.3}$$

In local coordinates, if f and $f' = Mf$ are regular generators of $\mathcal{I}$ on two overlapping charts, the Koszul generators are related by

$$\mathrm{d}\varepsilon'_i = \sum_j \bar{M}_{ij}\, \mathrm{d}\varepsilon_j, \tag{4.4}$$

the transition rule of $\mathcal{I}/\mathcal{I}^2$ in the bases $\bar{f}'$, $\bar{f}$.

Proof. $\mathbb{L}_{Z/X}$ is a globally defined complex of sheaves (§2.4, item 5). Locally it is concentrated in degree 1 by (4.1b), so $\mathbb{L}_{Z/X} \simeq H_1(\mathbb{L}_{Z/X})[1]$, and $H_1(\mathbb{L}_{Z/X}) \simeq \mathcal{I}/\mathcal{I}^2$ by Proposition 4.1, an isomorphism compatible with localization and hence an isomorphism of sheaves. By Lemma 4.2, on each chart it reads $\mathrm{d}\varepsilon_i \mapsto \bar{f}_i$; since $\bar{f}'_i = \sum_j \bar{M}_{ij} \bar{f}_j$ in $\mathcal{I}/\mathcal{I}^2$, (4.4) follows. □

Remark 4.4 (the gluing is computed, not chosen)**.** The gluing of $\mathbb{L}_{Z/X}$ is fixed by its definition (functoriality of the canonical resolution, or equivalently (2.4)) and makes no reference to f. Corollary 4.3 *computes* it. The same rule can also be obtained by comparing resolutions directly: $\varphi : K(f') \to K(f)$, $\varepsilon'_i \mapsto \sum_j M_{ij}\varepsilon_j$, commutes with d because $f' = Mf$, and is a map over A/I; by uniqueness of maps between resolutions up to homotopy, $\mathbb{L}(\varphi)$ is the comparison map, which is (4.4). The matrix M is not unique, but two choices differ by syzygies of f, which are Koszul relations with entries in I; hence $\bar{M}$ is unique and the cocycle condition holds on the nose.

Remark 4.5 (contrast: the derived self-intersection)**.** The analogous statement fails for $\mathcal{O}_Z \otimes^{\mathbb{L}}_{\mathcal{O}_X} \mathcal{O}_Z$. Locally it is $\bigoplus_i \bigwedge^i N^\vee[i]$ (formal), and its homology sheaves $\mathrm{Tor}_i = \bigwedge^i N^\vee$ are canonical; yet globally $\mathcal{O}_Z \otimes^{\mathbb{L}}_{\mathcal{O}_X} \mathcal{O}_Z \simeq \bigoplus_i \bigwedge^i N^\vee[i]$ holds if and only if $N_{Z/X}$ extends to the first-order neighborhood of Z [AC12]. A complex with homology in several degrees carries global extension data between them, invisible locally. $\mathbb{L}_{Z/X}$ escapes this because regularity concentrates it in a single degree. For the diagonal $Z = \Delta \subset X \times X$ the obstruction vanishes, which is why the HKR isomorphism holds globally.

4.4 Where regularity enters: a non-example

Let $A = k[x, y]$ and $I = (x^2, xy)$, so that $B = A/I$ is the line $\{x = 0\}$ with an embedded point at the origin. The Koszul complex of (x^2, xy) is not acyclic: the cycle $c = y\varepsilon_1 - x\varepsilon_2$ (note $dc = yx^2 - x \cdot xy = 0$) is not a boundary, since $d(\varepsilon_1\varepsilon_2) = x^2\varepsilon_2 - xy\varepsilon_1 = -xc$, and

$$H_1(K) = A\,c/A\,xc \simeq k[y] \neq 0. \tag{4.5}$$

So K is not a resolution of B; the formula (4.1b) would compute the cotangent complex of the derived zero locus $\operatorname{Spec} K$, not of B. Following Tate, adjoin a generator U of degree 2 with $dU = c$. In Ω^1 this gives $\mathrm{d}(dU) = y\,\mathrm{d}\varepsilon_1 - x\,\mathrm{d}\varepsilon_2$ (as $\mathrm{d}x = \mathrm{d}y = 0$ over A), and after $\otimes B$,

$$\cdots \to B\,\mathrm{d}U \xrightarrow{\mathrm{d}U \mapsto \bar{y}\,\mathrm{d}\varepsilon_1 - \bar{x}\,\mathrm{d}\varepsilon_2} B\,\mathrm{d}\varepsilon_1 \oplus B\,\mathrm{d}\varepsilon_2 \to 0. \tag{4.6}$$

- H_1 is B^2 modulo the relation $\bar{y}\,\mathrm{d}\varepsilon_1 = \bar{x}\,\mathrm{d}\varepsilon_2$, which is I/I^2 with its relation $y \cdot \overline{x^2} = x \cdot \overline{xy}$, as Proposition 4.1 requires.
- The map $B^2 \to I/I^2$, $\mathrm{d}\varepsilon_i \mapsto \bar{f}_i$, is no longer injective; its kernel is the image of $\mathrm{d}U$. This is the failure of $(A/I)^c \simeq I/I^2$ without regularity.
- Further generators are needed in degrees ≥ 2, and $\mathbb{L}_{B/A}$ is not bounded: by Avramov's theorem (Quillen's conjecture), $H_n(\mathbb{L}) = 0$ for $n \gg 0$ if and only if B is a local complete intersection [Avr99], and B is not.
- Since $A = k[x, y]$ is smooth over k, the transitivity triangle (2.6) for $k \to A \to B$ gives $H_n(\mathbb{L}_{B/k}) \simeq H_n(\mathbb{L}_{B/A})$ for $n \geq 2$, while H_0, H_1 of $\mathbb{L}_{B/k}$ are computed by the two-term complex (5.2). So the absolute cotangent complex is unbounded as well, and (5.2) only computes $\tau_{\leq 1}\mathbb{L}_{B/k}$.

5 The absolute cotangent complex and singularities

5.1 The two-term complex

Let P be smooth over k (e.g. a polynomial ring) and $B = P/I$. The transitivity triangle (2.6) for $k \to P \to B$ reads

$$\Omega_{P/k} \otimes_P B \to \mathbb{L}_{B/k} \to \mathbb{L}_{B/P} \xrightarrow{+1}. \tag{5.1}$$

If I is generated by a regular sequence, $\mathbb{L}_{B/P} = I/I^2[1]$ by Corollary 4.3, and

$$\mathbb{L}_{B/k} \simeq \left[I/I^2 \xrightarrow{\bar{f} \mapsto \mathrm{d}f \otimes 1} \Omega_{P/k} \otimes_P B \right] \qquad \text{(degrees } 1, 0\text{)}. \tag{5.2}$$

In general (5.2) only computes $\tau_{\leq 1}\mathbb{L}_{B/k}$. Three cases:

- *A point in* $\mathbb{A}^n$. The map is $\bar{x}_i \mapsto \mathrm{d}x_i$, an isomorphism: $\mathbb{L}_{k/k} = 0$.
- *A smooth subvariety* $Z \subset X$. Then $\mathbb{L}_{Z/k} = \Omega_Z$, and (5.2) says that the conormal sequence $0 \to N^\vee_{Z/X} \to \Omega_X|_Z \to \Omega_Z \to 0$ is exact, including on the left.
- *A singular* B. The map need not be injective, and $H_1(\mathbb{L}_{B/k}) \neq 0$ then shows that B is not smooth. The simplest case follows.

5.2 The fat point: what Ω^1 misses

Let $B = k[x]/(x^2)$ with $\operatorname{char} k \neq 2$, $P = k[x]$, $I = (x^2)$ (regular). By (5.2),

$$\mathbb{L}_{B/k} = \left[B\, e \xrightarrow{d} B\, \mathrm{d}x \right], \qquad d(e) = \mathrm{d}(x^2) = 2x\, \mathrm{d}x. \tag{5.3}$$

- $H_0(\mathbb{L}) = \Omega^1_B = B\,\mathrm{d}x/(2x\,\mathrm{d}x) = k\,\mathrm{d}x$ and $H_1(\mathbb{L}) = \ker d = k \cdot xe \neq 0$.
- First-order symmetries: $T^0 = \operatorname{Hom}_B(\Omega^1_B, B) = k \cdot x\partial_x$, the same from Ω^1 or from $\mathbb{L}$.
- First-order deformations: $T^1 = H^1\big(\operatorname{RHom}_B(\mathbb{L}, B)\big) = \operatorname{coker}\big(B \xrightarrow{a \mapsto 2xa} B\big) = B/(x) = k$. By (2.5) this is $\operatorname{Exalcomm}_k(B, B)$: the deformation $x^2 = t$, in which the double point splits into two.

By contrast, B is self-injective (a Frobenius algebra), so $\operatorname{Ext}^{\geq 1}_B(\Omega^1_B, B) = 0$: a deformation theory built from Ω^1 alone would conclude that B is rigid. Concretely, perturbing the equation to $x^2 + t\,g(x)$, the part of g absorbed by $x \mapsto x + ta(x)$ is $2xa$, and what remains is $g \bmod (2x)$, i.e. the constant term: this is $\operatorname{coker}(a \mapsto 2xa)$ again. For a general hypersurface $P/(f)$ the same computation gives $T^1 = P/(f, \partial f)$, the Tjurina algebra.

	Ω^1 only	cotangent complex $\mathbb{L}$
first-order symmetries T^0	$k \cdot x\partial_x$	$k \cdot x\partial_x$
first-order deformations	$\operatorname{Ext}^1(\Omega^1, B) = 0$ (wrong)	$T^1 = k$: $x^2 = t$
detects the singularity	yes: $\Omega^1_B = k\,\mathrm{d}x$ is not projective	yes; $H_1(\mathbb{L}) \neq 0$

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